

Beyond Silver: Comparative Performance Analysis of Alternative Metallization in Silicon Solar Cells

Veysel Unsur[✉], Melisa Korkmaz Arslan[✉], Seda Kilickaya Unver[✉], Esma Alloji[✉], and Rasit Turan

Abstract—The reliance of photovoltaic industry on silver (Ag) for silicon solar cell metallization faces critical challenges due to Ag scarcity, cost volatility, and sustainability concerns. Here, we systematically compare advanced metallization strategies that minimize Ag usage while preserving cell performance. We evaluate different approaches such as ultrafine screen printing of Ag, hybrid core-shell metal pastes, pure Ag-free metal pastes (Cu, Ni, Al/Si), and novel deposition techniques of electroplating and multinozzle dispensing. Fine-line printing enables Ag finger widths as narrow as 10–15 μm , reducing Ag consumption by up to 30% without compromising efficiency. However, fingers widths below 10 μm introduce printability challenges and increased series resistance. Hybrid core-shell pastes, such as Cu-core/Ag-shell (Ag/Cu) configurations, mitigate oxidation of Cu and diffusion risks at low temperatures but face limitations in high-temperature processes. To address this, pure Cu and Ni contacts demonstrate a breakthrough, achieving power conversion efficiencies comparable to traditional Ag contacts while slashing Ag usage to almost zero. Meanwhile, Al/Si pastes, while limited by lower voltage, show promise for rear-side metallization if used with Al/(boron doped) Si or Al/Ge alloy mixtures. These comparative findings highlight the trade-offs among Ag-reduction approaches and delineate a clear path toward significantly lower-Ag, high-efficiency silicon solar cells.

Index Terms—Copper (Cu) contacts, interdigitated back contact (IBC), metallization, nickel (Ni) contacts, photovoltaics (PV), screen printing, silver (Ag)-free solar cells, tunnel oxide passivated contact (TOPCon).

I. INTRODUCTION

THE growth of the photovoltaic (PV) sector to meet global decarbonization goals has dramatically increased the demand for silicon solar modules and with it the demand for silver (Ag) as the primary metallization material. Current projections suggest that by 2030 the PV industry could consume an amount of silver equivalent to roughly 40% of the world's annual supply [1]. If high-efficiency, Ag-intensive technologies such as tunnel oxide passivated contact (TOPCon) and silicon heterojunction (SHJ) solar cells dominate, Ag demand could soar to exhaust

up to 85%–98% of known reserves by mid-century [1], [2]. Traditional screen-printed Ag contacts consuming $\sim 12 \text{ mg/W}_p$ of Ag for TOPCon solar cells have fueled this demand, consistent with recent industry data from the 2025 ITRPV report [3], but the combination of limited global supply and all time high prices now threaten PV scalability and cost targets.

To sustain the terawatt-scale growth of PV, the industry is pursuing aggressive silver reduction and replacement strategies. One proven approach is to dramatically narrow the front-side contact fingers to reduce Ag usage. Schweigstill et al. [4], has demonstrated ultrafine Ag grid lines shrinking from $\sim 120 \mu\text{m}$ in 2005 to below $\sim 20 \mu\text{m}$ and even $\sim 10 \mu\text{m}$ in mass production [5]. Likewise, experiments on SHJ cells have shown that 20 μm -wide silver fingers can be printed while still maintaining low grid resistance [6]. These ultranarrow fingers, combined with multiwire interconnects in place of conventional soldered ribbons, allow higher series resistance tolerances and reduce module shading losses.

While ongoing process and design optimizations can stretch every milligram of silver, achieving sustainable terawatt-scale deployment will require reducing silver consumption below $\sim 2 \text{ mg/W}_p$, which is a threshold identified in recent studies as necessary to prevent resource constraints in the coming decades [7], [8]. This does not imply the complete elimination of silver, but rather the development of metallization strategies that drastically minimize its usage without compromising performance. Copper (Cu) is considered the most promising alternative, owing to its electrical conductivity of $\sim 6 \cdot 10^7 \text{ S/m}$, which is only slightly lower than the Ag of $\sim 6.3 \cdot 10^7 \text{ S/m}$, and its significantly greater abundance with an estimated global reserve of over 870 Mt compared to $\sim 530\,000 \text{ t}$ of Ag [9]. Recent studies confirm that Cu-based front contacts can match Ag-based performance when properly engineered [10]. In heterojunction (SHJ) cells, the intrinsic -transparent conductive oxide (TCO) layers act as diffusion barriers and enable screen-printing of Cu or Ag/Cu pastes with nearly the same efficiency as all-silver metallization [11]. However, Cu has well-known challenges in which it oxidizes readily during high-temperature firing in air [12] and can diffuse into silicon to form deep-level defects [13], [14], [15]. Thus, any Cu paste used for high-temperature applications as in TOPCon and interdigitated back contact (IBC) technologies must include protective measures: dopants or glass additives to retard oxidation, and/or metallic barrier layers to prevent Cu–Si reactions. Other metals have also been explored. Nickel (Ni) offers excellent resistance to corrosion [16] and can serve as a pure Ni contact, but its volume resistivity is higher than Ag and

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Veysel Unsur is with the ODTU-GUNAM, Middle East Technical University, 06800 Ankara, Türkiye, and also with the Department of Electrical and Electronics Engineering, Necmettin Erbakan University, 42090 Konya, Türkiye (e-mail: veysel.unsur@odtugunam.org).

Melisa Korkmaz Arslan, Seda Kilickaya Unver, and Rasit Turan are with the ODTU-GUNAM, Middle East Technical University, 06800 Ankara, Türkiye.

Esma Alloji is with the Department of Nanoscience and Nano Engineering, Necmettin Erbakan University, 42090 Meram, Türkiye.

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screen-printed Ni contacts on the front side of TOPCon solar cells tends to yield higher line and contact resistances due to higher metal work function of Ni [17]. To solve this, Unsur et al. [18], demonstrated that 4% Ag doping on the Ni particles can lower both contact and line resistance and show similar results compared to Ag counterparts [19], [20].

To capitalize on the strengths of different materials, hybrid and core-shell metallization pastes have been proposed. Cu-core/Ag-shell (AgCu) particles combine the low cost and high conductivity of copper with a thin silver coating that prevents oxidation and facilitates stable contact formation. These pastes have shown promising electrical properties; for example, a boron-doped AgCu core-shell paste sintered in air at up to 800°C with a boron glass additive produced a conductive copper-rich film with low resistivity [21]. However, sintering such structures at high temperatures can lead to local melting or rupture of the thin Ag shell, causing copper penetration and diffusion into the silicon, which poses reliability risks. As a result, core-shell pastes are more commonly used in low-temperature architectures such as SHJ, where the intrinsic TCO layer acts as a diffusion barrier, enabling stable Cu integration and demonstrating good performance [22]. For high-temperature cell concepts such as TOPCon and IBC, an emerging strategy is the dual-printing approach: first printing a fine-line Ag grid using a conventional high-temperature paste to form the contact interface, followed by a low-temperature printed AgCu paste that provides bulk conductivity [23].

While several studies have investigated individual metallization strategies such as ultrafine silver printing, Cu-based pastes, or plating techniques, a number of recent review works have also provided broad overviews of metallization technologies for high-efficiency silicon solar cells [24], [25]. However, a systematic, side-by-side comparison that quantitatively evaluates the trade-offs in Ag consumption, electrical performance, process compatibility, and industrial maturity across architectures such as SHJ, TOPCon, and IBC remains limited. In particular, comparative assessments that emphasize silver reduction and replacement strategies within screen-print-compatible and industrially relevant frameworks are still scarce. This work aims to fill that gap by providing a technically grounded analysis of metallization pathways with a clear focus on practical Ag-saving solutions for advanced silicon solar cell technologies.

II. STRATEGIES AND OUTCOMES FOR AG REDUCTION AND AG FREE SOLAR CELLS

A. Ag Reduction Through Fine-Line Printing

Early industrial Si solar cells employed screen-printed front-side fingers approximately 120–140 μm [26]. Such wide electrodes caused significant optical shading corresponding to over 5% shading on the front surface and high Ag consumption of around 200 mg/cell [25], [27], limiting efficiency and raising metallization cost. Over the past decade, a remarkable shrinkage in finger widths from $\sim 120 \mu\text{m}$ to under 20 μm was achieved by laser enhanced contact optimization (LECO) [28], higher meshed screens [29], quality pastes enabling dramatic reductions in front-side shading and Ag usage [30], [31], [32]

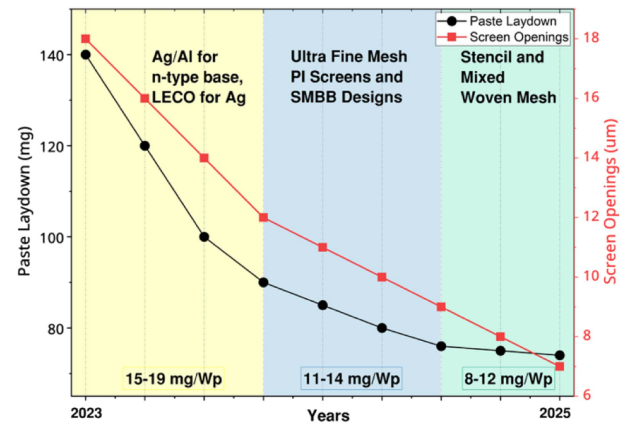


Fig. 1. Evolution of Ag paste laydown over time along with the main cause of the development.

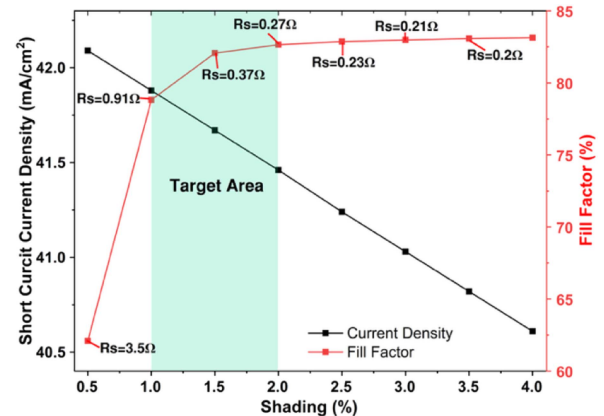


Fig. 2. Simulated J_{SC} and FF response of an M10-size TOPCon solar cell as a function of front-side shading, calculated using Griddler.

and lastly the evolution of new cell interconnection concepts (multibusbar, super multibusbar) [33], [34], [35] as illustrated in Fig. 1. This evolution has been crucial for enabling conventional PERC cells to exceed 20% efficiency and remains even more critical in high-efficiency architectures such as TOPCon, SHJ, and IBC, where achieving an optimal trade-off between shading, resistive, and recombination losses is essential. Ultrafine fingers play a key role in this optimization, enabling Ag laydown to be minimized without sacrificing electrical performance.

The transition to multibusbar cells illustrated the impact of finer lines and optimized grids [34], [35]. Increasing busbar count allowed using more, narrower fingers without raising resistive losses, thereby cutting shading and boosting current [25]. Through incremental improvements such as dual-printing techniques [36], [37] and advancements in screen technology including the use of higher mesh counts, reduced wire diameters, and optimized emulsion profiles, screen-printed finger widths have dropped below 20 μm . This continuous refinement delivered measurable performance gains in which finer lines directly increase the current by reducing shading and can improve V_{OC} by lowering the metal fraction and associated recombination.

Fig. 2 obtained from Griddler [38] simulations of a standard M10-size TOPCon solar cell, in which all cell parameters were

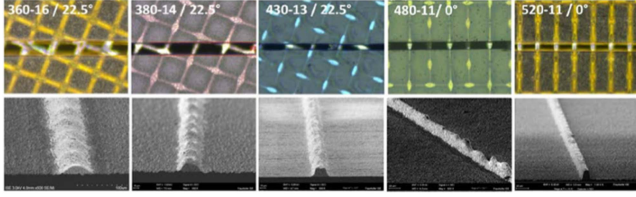


Fig. 3. Evolution of screens for fine line screen printing [42].

held constant except for the front metallization layout. The simulations indicate that reducing front-side shading from 4% to $\sim 1.5\%$ through optimization of the finger width and number of busbars results in a $1.2\%_{\text{abs}}$ efficiency increase, driven by improvements in current density (J_{SC}) and fill factor (FF).

A major driver of finer fingers has been the evolution of screens used in screen printing as shown in Fig. 3. Traditional screens used stainless steel meshes of ~ 360 wires/inch with various wire diameters and emulsion-over-mesh (EOM) thickness [25]. Such screens produced wide finger openings and thick mesh marks in the printed lines. Modern high-performance screens for fine line printing use 520 to 750 wires/inch with ultra-fine wire diameters down to $4\ \mu\text{m}$ with etch-back utilizing high-tensile materials [39]. The capability to print fingers below $20\ \mu\text{m}$ depends not only on the mesh count and emulsion design, but also critically on screen durability and paste release dynamics. While stainless steel meshes have traditionally dominated, recent developments have enabled ultra-fine wire diameters using advanced etched back stainless steel screens. A key breakthrough has been the introduction of polyimide barrier layers, which significantly improve screen mechanical stability and extend printing lifetimes to 500,000 to 1,000,000 prints per screen [40]. Equally critical has been the adoption of knotless mesh technologies [41], aligning mesh wires with finger orientation to eliminate crossover knots and improve paste flow uniformity. This results in significantly improved finger edge definition, reduced line width variation, and better paste release enabling consistent printing of sub- $20\ \mu\text{m}$ fingers at industrial scale.

Concurrently printable finger aspect ratio is influenced by multiple factors, including the screen channel geometry (EOM and sidewall angle), paste rheology, and printing parameters. While early fine-line screens relied on increased EOM to achieve taller fingers, modern screen designs have shifted toward reduced EOM values as low as $2\text{--}3\ \mu\text{m}$ to minimize silver consumption [43]. The mesh configuration has also evolved: whereas 22.5° mesh angles were once standard, today's knotless 0° mesh screens are widely adopted for front-side metallization. Nonetheless, continuous improvements in mesh metallurgy, calendaring (to flatten wires and expand open area) and development of laser-structured PI barrier layers are incrementally closing the gap to the ultranarrow regime [40]. As screen printing pushes toward the $\sim 10\text{--}15\ \mu\text{m}$ linewidth barrier, fundamental mesh limitations (minimum wire diameter, mesh stability) begin to constrain further progress. Thus, stencil printing is gaining attention as a promising workaround to achieve fingers below $10\ \mu\text{m}$ in an industrially viable way. In stencil printing, a metal

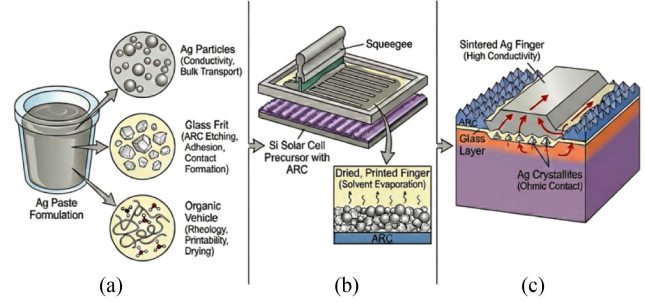


Fig. 4. Metal paste formulation, application, and resulting contact structure (created with AI). (a) Paste Formulation & Components. (b) Printing & Drying Process. (c) Fired Contact Formation & Function.

foil (or other substrate) with patterned openings is used instead of a woven mesh, eliminating the issue of screen wires blocking narrow apertures [43], [4].

Printing ultranarrow fingers demands not only advanced screens but tailored metallization pastes as well. A typical screen-printable metallization paste comprises three main components: conductive metal particles that provide electrical conductivity, a glass frit that facilitates the etching of passivation layers, sintering and contact formation, and an organic vehicle that governs the paste's rheological behavior and printing characteristics [44], [45], [46], as illustrated in Fig. 4. The rheology of the paste is critical for fine-line printing. To penetrate $\sim 10\ \mu\text{m}$ screen openings, the paste must exhibit a sufficiently low viscosity under high shear rates in the range of $10\text{--}50\ \text{Pa}\cdot\text{s}$ at $10^3\ \text{s}^{-1}$, yet a high enough yield stress of $> 300\ \text{Pa}$ once deposited to prevent excessive spreading on the textured wafer surface [47]. Achieving this balance often involves careful adjustment of paste chemistry using optimally sized metal particles of median diameter (D50) of $1\text{--}2\ \mu\text{m}$ and thixotropic binders to ensure good aspect ratio of the contacts.

Overall, while ultrafine-line printing has significantly reduced silver usage through screen and stencil innovations, further gains are limited by mesh design, paste rheology, and printing resolution. To continue driving Ag consumption downward without compromising electrical performance, the industry is increasingly turning to material-level strategies.

B. Core / Shell and Hybrid Pastes for Ag Reduction

Hybrid metallization approaches combine Ag with lower cost Cu to reduce material costs while maintaining electrical performance. Although pure Cu offers excellent conductivity, it is prone to oxidation in air and diffuses readily into silicon, potentially degrading the junction and passivation quality [12], [15]. To address these challenges, screen-printable core-shell pastes have been developed in which Cu particles are coated with a thin Ag shell and submicron Ag particles are incorporated in the paste that fill the interstitial voids between larger Ag/Cu particles as evident in the SEM images seen in Fig. 5. These Ag/Cu particles provide a conductive network while offering limited protection against oxidation as the Ag shell typically exhibits pinholes and can degrade at high temperatures, rendering it ineffective as a standalone diffusion barrier. As such, this

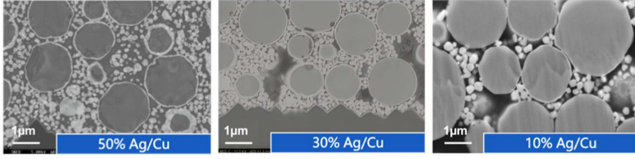


Fig. 5. SEM images of Ag coated Cu core/shell contacts with different Ag loading [35].

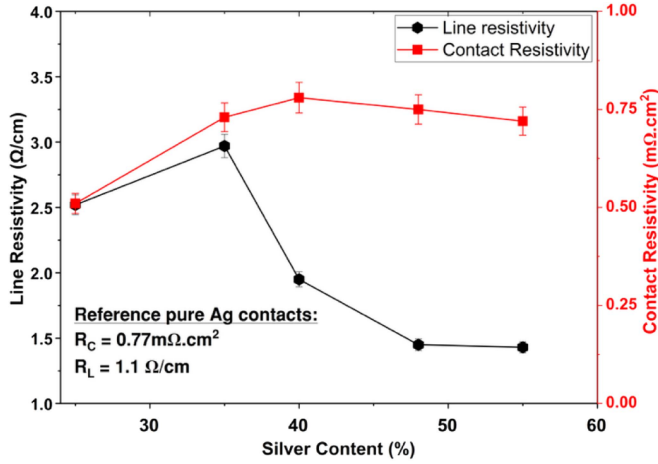


Fig. 6. TLM and line resistance measurement of industrial SHJ solar cells printed with Ag/Cu core shell paste with different loading. The data points show the mean values of 5 SHJ solar cells from each group.

approach is suitable primarily for cell architectures that include an intrinsic diffusion barrier, such as TCO layer in SHJ solar cells [22], [48].

Low-temperature cell architectures such as SHJ are especially well suited to Ag/Cu pastes. These cells require curing the metallization at only $\sim 200^\circ\text{C}$ (to avoid damaging the amorphous silicon layers), which is gentle on Ag/Cu particles. At such temperatures, the thin Ag shells can sinter together to form continuous conduction paths, while Cu cores remain somehow encapsulated. Moreover, SHJ cells include TCO films on the silicon; this TCO acts as an intrinsic barrier that can impede any Cu that might diffuse out of the paste. The presence of a TCO buffer means a Cu-based finger does not directly touch the silicon active layers, greatly alleviating diffusion concerns. This intrinsic passivation of the contact makes Ag/Cu designs particularly attractive for SHJ.

Fig. 6 illustrates the key contact parameters of contact (R_C) and line resistivities (R_L) on SHJ solar cells. Commercially obtained unmetallized commercial blue cells were printed using commercially available pastes with varying Ag content and subsequently cured at 220°C for 4 min. The R_L increases with decreasing Ag content, reaching a peak of $\sim 3 \text{ }\Omega/\text{cm}$ at $\sim 35 \text{ wt}\%$ Ag. This trend is attributed to the reduced sintering quality between core-shell particles compared to pure Ag systems, which impedes lateral conductivity.

However, at 25 wt% Ag, the R_L decreases slightly, which may result from improved particle packing or paste formulation differences specific to that composition. On the other hand, R_C

TABLE I
Comparative I-V Measurements of SHJ Solar Cells Printed With Pure Ag and Ag/Cu Core Shell Contacts

	η (%)	J_{SC} (mA/cm^2)	V_{OC} (mV)	FF (%)	R_C ($\text{m}\Omega/\text{cm}^2$)	R_L (Ω/cm)
Ag ₂₅ /Cu ₇₅	22.69	39.41	718	80.2	0.51	2.52
Pure Ag	22.94	39.38	721	80.8	0.77	1.14

values remain stable and comparable to pure Ag counterparts of $0.77 \text{ m}\Omega/\text{cm}^2$, confirming that low-temperature-cured Ag/Cu pastes can maintain vertical contact formation on SHJ cells. While the electrical contact is governed by the Ag-rich phase forming the interface with the semiconductor, the R_C does not scale monotonically with silver content. At intermediate Ag fractions, microstructural changes such as altered particle packing and reduced interfacial conformity can lead to a temporary increase in R_C , before a continuous Ag network is re-established at higher Ag loadings.

Table I shows the I-V electrical output parameters of the printed SHJ solar cells with 25% Ag loading on 75% Cu paste (Ag₂₅Cu₇₅) in which the slight decrease due to the higher line resistance of Ag/Cu core shell contacts occurred. This is due to the fact that volume resistance of Cu powders is higher than Ag counterparts and sintering is not as effective as Ag contacts. Although the R_L remains higher than in pure Ag pastes, the printed Ag/Cu fingers achieved functionally sufficient conductivity at 25% Ag loading, thanks to bulk conduction of densely packed Cu cores. In addition, the inclusion of submicron Ag particles in the paste formulation helps bridge particle gaps and enhances sintering at low temperatures, partially compensating for the reduced lateral conductivity inherent to core-shell structures. Importantly, these Ag/Cu contacts demonstrated excellent stability where resisted oxidation up to $\sim 300^\circ\text{C}$ in TGA testing, and the printed fingers passed extended thermal aging (150°C for 240 min) without additional performance degradation compared to Ag fingers.

Beyond SHJ cells, hybrid metallization strategies are being investigated for other advanced architectures such as TOPCon and IBC [23], [49]. Implementing Cu-based pastes in these contexts is more challenging because they typically involve high-temperature steps up to $\sim 800^\circ\text{C}$. At such temperatures, even Ag/Cu particles risk oxidation in which the Ag shell can break or de-wet from the Cu and could allow Cu diffusion into silicon. One strategy to further reduce silver consumption in TOPCon and IBC solar cells is the use of floating busbars, which are printed on insulating passivation layers such as SiN_x . Unlike conventional pastes, the floating busbar pastes lack glass frits and therefore do not etch through the passivation layer, making them electrically noncontacting. In this configuration, the busbars serve solely as conductive pathways for lateral current collection. This allows the use of low-cost Ag/Cu or pure Cu-based pastes for the noncontacting segments, significantly reducing the silver content while maintaining overall conductivity. Such approaches are particularly relevant for high-efficiency architectures where the contacting grid and current-carrying layers can be decoupled [50]. Chen et al., [49] demonstrated

that screen-printing Cu paste busbars on an IBC cell with only the fine fingers using Ag yielded no efficiency loss compared to all-Ag metallization. The IBC cells achieved $\sim 22.7\%$ efficiency with Cu busbars, only $\sim 0.1\%$ lower than the 22.8% reference, while cutting total Ag usage roughly in half to $\sim 4.5 \text{ mg/W}_p$. Furthermore, Zhang et al., showed that same “floating” concept can be applied to fingers on the backside of a TOPCon solar cell as well [23]. By printing commercial Ag paste on a pattern where the Ag underneath corresponds to 1% coverage and Cu full contacts on top of Ag contact layers can lead to 85% of Ag reduction without losing significant efficiency [37].

The key takeaway is that hybrid Ag/Cu metallization need not be monolithic where it can involve spatially distinct materials (Ag for contacts, Cu for busbars/fingers) or composite particles, as long as the overall design ensures electrical continuity and stability. Therefore, recent prototypes on SHJ, TOPCon, and IBC indicate that Ag consumption can be reduced by up to 80% via Ag/Cu paste solutions with only minor efficiency losses. While early studies have shown comparable electrical performance, the long-term reliability of Ag/Cu metallization remains a critical area for further validation in module-integrated environments.

C. Alternative Ag-Free Conductive Materials

Although reducing Ag through fine line printing and hybrid metallization yields promising results, eliminating silver completely in solar cell metallization can dramatically lower costs, however, any alternative must offer high conductivity and preserve junction passivation quality. Recent work has focused on screen-printable pastes that completely avoid Ag, including pure Cu [51], Ni [18], and aluminum/silicon (Al/Si) formulations [52]. These Ag-free conductors are engineered for compatibility with high-efficiency architectures such as TOPCon, often requiring tailored firing profiles and LECO techniques to overcome challenges such as oxidation, interdiffusion, and thermal expansion mismatch.

High-temperature pure Cu pastes for TOPCon and IBC have emerged as promising replacements for screen-printed Ag, provided they can be processed in air without oxidizing and without copper diffusing into silicon. One notable example is Cu paste by Bert ThinFilm, a front-side Cu paste formulated to etch through the SiN_x antireflection coating (ARC) and fire in air at peak $\sim 600^\circ\text{C}$ [51], [53]. The formulation is designed to prevent copper oxidation during firing and constrains Cu diffusion into the silicon by forming a metal oxide-based glass layer, which originates from the glass frit composition and remains as a residual glass layer after firing. This strategy mitigates Cu diffusion into the emitter, preserving passivation quality. In a recent demonstration on the front side of a commercial n-TOPCon solar cell, the Cu-contacted solar cell achieved over 24% efficiency with an open-circuit voltage equivalent to conventional Ag counterparts. Despite these encouraging results, key challenges remain unresolved, notably, maintaining oxidation resistance and preventing Cu diffusion into silicon during high-temperature firing. These issues persist due to the potential incompleteness or local breakdown of the glass frit-originated barrier layer. Tailored paste formulations, incorporating optimized organic vehicles and glass frit chemistries, may aim to mitigate these risks

TABLE II
Electrical Performance of Ni Contacted Solar Cells

	η (%)	J_{sc} (mA/cm^2)	V_{oc} (mV)	FF (%)	R_c ($\text{m}\Omega/\text{cm}^2$)	R_L (Ω/cm)
Ni contact	23.6	41.21	705	81.2	0.51	1.92
Ag/Al ref	23.7	41.15	708	81.6	0.77	1.14

by forming more uniform and adherent oxide-based diffusion barriers during firing.

Another Ag-free alternative metal is Ni that has long been considered as a lower cost metallization and it forms stable nickel silicide contacts [54]. However, directly screen-printing Ni onto an Si emitter poses challenges such as pure Ni pastes fired in air tend to oxidize and have poor adhesion and high-contact resistance [55]. A recent study of ours [18] demonstrated that a screen-printed Ni paste can achieve efficiency parity with Ag on industrial n-TOPCon cells. The paste is a high-temperature fire-through Ni formulation featuring an engineered glass frit to impart hole-selective contact properties. After screen-printing on the boron-diffused front emitter of an n-TOPCon cell, the Ni contacts were cofired in a belt furnace with a peak temperature around 810°C , similar to conventional Ag paste firing. As shown in Table II [18], Ni contacts achieved 23.6% efficiency closely similar to the 23.7% efficiency of reference cells with standard Ag/Al paste. This demonstrates that screen-printed Ni contacts can fully rival silver contacts on n-TOPCon cells. The small Ag addition is a practical trade-off reducing silver consumption by $\sim 98\%$ while avoiding the pitfalls of pure Ni, namely, the tendency of Ni to oxidize during firing, its higher metal work function which can result in elevated contact resistance, and the lack of NiSi formation that otherwise facilitates low-resistance ohmic contact. Notably, the function of the glass frit as a local passivating contact is a novel concept to mitigate metal-silicon interface recombination.

Finally, aluminum pastes are widely used on the rear side of p-type PERC cells to form the back-surface field (BSF), but using Al for contacts on advanced cell architectures (especially n-type devices) requires overcoming Al-Si alloying effects. Al/Si conductive pastes, which incorporate silicon powder into the Al paste, have been investigated as an Ag-free alternative for contacting n-type TOPCon [56] and also p-TOPCon [57] structures. Aluminum is an attractive conductor due to its much lower cost than Ag and suitable work function of $\sim 4.35\text{eV}$ for contacting n^+ /poly-Si. However, firing Al paste directly on an n^+ /poly-Si contact can induce excessive Al-Si alloy formation, locally converting the n-type region to p-type (an Al-doped layer analogous to a BSF) [58]. This unintended p^+ region creates a built-in field opposing the junction, thereby reducing carrier collection and eroding V_{oc} . In addition, heavy alloying produces voids and spikes that can degrade the passivation and mechanical integrity at the contact [59]. The challenge is thus to enable Al paste contacts with limited alloy depth and minimal passivation damage.

Ding et al. [56] reported that screen-printing Al pastes with various Si weight percentages onto the rear n^+ /poly-Si layer of n-TOPCon cells. Instead of using glass frit in the paste itself, laser contact opening (LCO) was utilized to expose the

polySi, then printed Al paste containing 1–29 wt% Si and cofired the cells (peak firing temperature in the $\sim 800^\circ\text{C}$ range, similar to Ag firing). Higher Si content in the paste was found to suppress the thickness of the Al-Si alloy layer and reduce void formation. In these formulations, the Si in the paste is lightly doped with boron or germanium to moderate the sintering kinetics, which is particularly important for p-type emitters, and enhance the electrical contact at the metal/semiconductor contact. The optimal composition was 25–29 wt% Si, which yielded a maximum cell efficiency of $\sim 22.5\%$, compared to $\sim 24.5\%$ for Ag-metallized counterparts. While the absolute efficiency was lower, this difference is largely attributed to limited Al/Si alloy formation and laser-induced damage from the LCO process. Notably, the fill factor, a more direct indicator of contact quality, was only 0.8% lower than the Ag reference, reflecting increased contact and line resistances for the Al/Si metallization. This small difference underscores the potential of Al/Si pastes as a viable Ag-free alternative with comparable contact performance.

To better assess the contact behavior, Al-Si pastes obtained from TOYAL were evaluated experimentally on the rear side of commercial M10 n-type industrial TOPCon solar cells. Local contact openings were formed using a 532 nm ns laser, followed by screen printing of two Al-Si pastes (Al-Si 1 and Al-Si 2, with ~ 15 wt% and ~ 40 wt% Si, respectively) and a reference Ag paste. The cells were cofired at peak temperatures ranging from 660°C to 820°C in 20°C steps. The relatively high contact resistivity can be decreased below $0.6 \text{ m}\Omega\cdot\text{cm}^2$ with firing optimization, as shown in Fig. 7. In Al-Si 1, the lowest contact resistivity of $0.53 \text{ m}\Omega\cdot\text{cm}^2$ was achieved at 720°C . Beyond 720°C , the R_C increased steadily with increasing temperature. A similar trend was observed for Al-Si 2, which contains a higher Si fraction; the R_C reached $1.4 \text{ m}\Omega\cdot\text{cm}^2$ at 740°C and remained in a comparable range up to 800°C .

While the above approach demonstrates the viability of Al/Si pastes, the efficiency penalty from the alloy-induced V_{OC} loss remains a concern. A more advanced strategy is to use a diffusion barrier layer between the Al paste and the Si [60]. A sputtered TiN_x layer (~ 35 nm) is introduced as a conductive diffusion barrier for Ag-free contacting of polySi passivating contacts. In their design, an n-type wafer with a full-area rear n-polySi/ SiO_x passivating contact was first coated with a thin TiN_x film (with an optional ~ 7 nm TiO_x seed layer to adjust crystal orientation) prior to screen-printing an Al paste on the rear. During firing (770 – 800°C in air), the TiN_x layer prevents Al and Si from interdiffusing, so no Al-Si alloy forms in the silicon and the passivation quality is preserved. With this TiN_x barrier layer, metallization induced voltage loss was kept under 10 mV even for structures with high initial implied V_{OC} (iV_{OC}) of > 720 mV. The barrier-enabled contacts showed iV_{OC} of > 710 mV and $< 3 \text{ m}\Omega\cdot\text{cm}^2$ of contact resistivity.

In conclusion, Al-based pastes with high Si content can partially replace Ag contacts (with some efficiency loss), but pairing standard Al pastes with a $\text{TiN}_x/\text{TiO}_x$ diffusion barrier is an even more promising route to achieve Ag-free metallization without sacrificing efficiency.

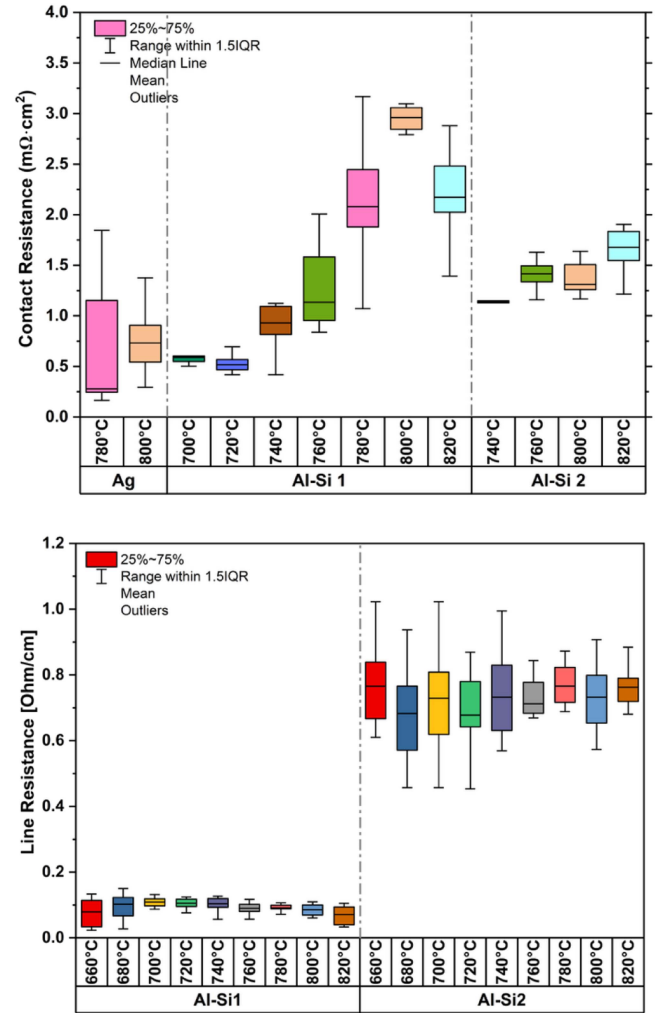


Fig. 7. (Top) Contact resistivity for different peak T for Ag, Al-Si, and (Bottom) Al-Si 2, and line resistivity for the Al-Si contacts.

III. ALTERNATIVE CONTACT FORMATION TECHNIQUES

In pursuit of finer contact grids and reduced silver usage, two promising alternatives to conventional screen-printed front contacts are being explored: 1) Ni/Cu/Ag electroplated metallization [61] and 2) multinozzle parallel dispensing [62]. These techniques enable ultrafine fingers below $\sim 20 \mu\text{m}$ [63] and high conductivity contacts and are being developed for industrial implementation on high-efficiency crystalline silicon cells.

Electroplating a Ni/Cu/Ag stack in place of screen-printed Ag paste is a well-established research/small-scale production pathway to lower metallization cost and improve efficiency [64], [65]. In this approach, a thin Ni layer is first plated (often after LCOs in the dielectric/passivation layer) to form a nickel silicide contact with silicon, followed by a thick copper plating for bulk conductivity, and a thin silver capping for solderability and corrosion protection. This method drastically reduces Ag consumption by substituting copper for most of the finger volume. It has been successfully demonstrated on a variety of cell types from p-type Al-BSF and PERC cells [66] to n-type designs indicating broad architectural compatibility [67], [68]. Notably,

plated Ni/Cu contacts have been applied to passivating-contact cells such as TOPCon and even IBC cells [69], which demand very fine feature metallization. Plating is considered an industrially feasible alternative to screen printing for next-generation cells, especially as it can contact lowly doped surfaces that are challenging for fired Ag pastes. While plating was traditionally favored for enabling ultrafine finger widths, above-mentioned advances in screen and stencil printing have narrowed this gap significantly. As such, the reduction of shading losses is no longer a unique advantage of plated metallization. However, plating still offers distinct benefits in terms of contact quality and conductor structure. For instance, the plated copper forms a dense, bulk metal contact rather than a porous sintered network, enabling excellent electrical conductivity and mechanical robustness. Copper can be selectively grown to several microns in thickness with high aspect ratios, helping to maintain low line resistance even with fine feature geometries.

Multinozzle parallel dispensing is an advanced printing approach that replaces the traditional screen and squeegee with an array of fine nozzles that continuously dispense silver paste lines [70], [71]. In essence, this is a noncontact printing technology where dozens of nozzles extrude conductive paste in parallel, directly onto the moving wafer surface. The goal is to achieve the same or better line precision as screen printing, but without a mesh screen thereby enabling ultrafine fingers, great uniformity and more efficient paste usage. Dispensing can produce ultrafine finger widths on the order of the nozzle orifice diameter, typically $\sim 20\text{--}30\text{ }\mu\text{m}$, with the latest developments pushing below $20\text{ }\mu\text{m}$ [70]. Because the paste is deposited as a continuous filament, the dispensed fingers have a very uniform cross-section and smooth sidewalls (no indents from screen mesh). Lorenz et al. [73] reported that their lab-scale parallel dispenser (15 nozzles of $\sim 22\text{ }\mu\text{m}$ each) printed TOPCon cell fingers with an average shading width of $\sim 23.5\text{ }\mu\text{m}$, compared to $\sim 26.4\text{ }\mu\text{m}$ with a state-of-the-art fine-line screen print, and some dispensed fingers were as narrow as $\sim 19\text{ }\mu\text{m}$ [72]. Dispensed lines also tend to exhibit higher aspect ratios for a given width, as the paste is deposited precisely without being forced through a mesh, thereby avoiding the distortions and edge roughness typically introduced by screen printing mesh marks.

Both Ni/Cu/Ag plating and multinozzle dispensing are advancing as viable alternatives to screen printing for front-side metallization of crystalline Si solar cells. Electroplating offers a route to ultrafine copper contacts with ultralow contact resistance and minimal silver, which is especially attractive for high-efficiency cells with passivating contacts (TOPCon, SHJ, IBC). Parallel dispensing, on the other hand, enables immediate gains in fine-line printing within the existing production paradigm using modified silver pastes and firing, but achieving narrower fingers and lower silver usage in an inline, high-throughput tool. Each technology comes with its own integration challenges: plating requires additional laser and plating steps (with chemical handling), whereas dispensing requires precision hardware to maintain consistent microprinting. Ongoing research and pilot demonstrations show that both approaches can deliver front contact fingers $\sim 20\text{ }\mu\text{m}$ width, contact resistivities on the order of $1\text{ m}\Omega\cdot\text{cm}^2$ or below, and cell efficiencies on par

with or better than screen-printed contacts. As the PV industry targets cell efficiencies beyond 26% and continues to reduce silver consumption, alternative metallization techniques such as plating and dispensing offer promising pathways for specific architectures or integration strategies. However, given the rapid advancements in screen and stencil printing, especially in achieving sub $10\text{ }\mu\text{m}$ features, these conventional methods are likely to remain the dominant industrial standard in the near term.

IV. CONCLUSION

This study provides a comprehensive comparison of emerging metallization strategies aimed at reducing or eliminating silver in silicon solar cells. Fine-line screen and stencil printing techniques enable narrower Ag fingers ($\sim 10\text{--}15\text{ }\mu\text{m}$), lowering Ag usage by up to 65% without sacrificing performance, though they require increased process precision and equipment upgrades.

Hybrid and core-shell pastes, such as Ag/Cu, offer a practical middle ground substantially reducing Ag content while maintaining electrical performance, particularly in SHJ cells. While their thermal limitations and synthesis complexity pose challenges for direct high-temperature firing, promising results have been demonstrated for TOPCon architectures using a double-layer metallization strategy, where a silver contact layer is combined with an AgCu conducting layer.

Ag-free approaches based on screen-printable Cu, Ni, and Al/Si pastes demonstrate compelling cost advantages and promising device efficiencies ($>22\%$). Cu pastes, in combination with engineered glass frits or diffusion barriers, have reached $>24\%$ efficiency in TOPCon cells. Ag-doped Ni pastes have achieved contact resistivities comparable to Ag and are compatible with standard firing profiles. Al/Si pastes, though limited by contact resistivity, remain promising for rear-contact applications when paired with barrier layers such as TiN_x .

Beyond printing, alternative contact formation techniques such as electroplating and multinozzle dispensing provide routes to ultrafine metal fingers and reduced silver consumption. Plated Ni/Cu/Ag stacks and high-throughput dispensing systems show industrial promise, though they require process integration and capital investment.

Cost-related considerations in this work are framed by the widespread industrial motivation to reduce reliance on silver, given its substantially higher raw material cost compared to Cu, Ni, or Al. The potential cost benefits discussed here assume that alternative metallization materials can reach similar levels of technological maturity, reliability, and paste formulation robustness. A quantitative cost assessment, including equipment, throughput, yield, and reliability factors, would require a dedicated techno-economic analysis and is therefore outside the scope of this study.

No single method universally meets all performance, cost, and manufacturability goals. The optimal pathway depends on cell architecture, production infrastructure, and silver reduction targets. Hybridizing material innovations with advanced deposition technologies will be critical to achieving high-efficiency, scalable, Ag-free PVs.

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